

Install Libraries

```
!pip install streamlit pyngrok pandas
```

```
Requirement already satisfied: streamlit in /usr/local/lib/python3.12/dist-packages (1.57.0)
Collecting pyngrok
  Downloading pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.4.1)
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<8,>=5.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (8.3.3)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/python3.12/dist-packages (from streamlit) (3.1.19)
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.0.2)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (26.1)
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (11.3.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.9.2)
Requirement already satisfied: protobuf<8,>=3.20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.12.0)
Requirement already satisfied: starlette>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.52.1)
Requirement already satisfied: uvicorn>=0.30.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.46.0)
Requirement already satisfied: httptools>=0.6.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.7.1)
Requirement already satisfied: anyio>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.13.0)
Requirement already satisfied: python-multipart>=0.0.10 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.0.26)
Requirement already satisfied: websockets>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous>=2.1.2 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.12/dist-packages (from pyngrok) (6.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (3.1.3)
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (1.42.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio>=4.0.0->streamlit) (3.13)
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.12)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (2.31.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (2025.11.12)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn>=0.30.0->streamlit) (0.16.0)
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,>=3.0.7->streamlit) (5.0.6)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (3.0.2)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (25.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.36.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.25.0)
Downloading pyngrok-8.1.2-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.2
```

Create Streamlit App

```
%writefile app.py

import streamlit as st
import pandas as pd
import random

# Load dataset
df = pd.read_csv("interview_questions.csv")

# Function to get random question
def get_question(role):

    filtered = df[df['role'] == role]

    row = filtered.sample(1).iloc[0]

    return row['question'], row['expected_keywords']

# Function to analyze answer
def analyze_answer(answer, keywords):

    score = 0
```

```

    answer = answer.lower()

    keyword_list = keywords.split(",")

    matched_keywords = []

    for word in keyword_list:

        if word.strip().lower() in answer:

            score += 2

            matched_keywords.append(word.strip())

    length_score = min(len(answer.split()) // 5, 4)

    total_score = score + length_score

    if total_score > 10:
        total_score = 10

    return total_score, matched_keywords

# Feedback function
def generate_feedback(score):

    if score >= 8:
        return "Excellent Answer! Very professional."

    elif score >= 5:
        return "Good Answer, but improve explanation."

    else:
        return "Try to explain with more technical detail."

# Streamlit UI
st.set_page_config(page_title="AI Interview Bot")

st.title("🤖 AI Interview Preparation Bot")

st.write("Practice technical interview questions using AI.")

# Role selection
role = st.selectbox(
    "Select Job Role",
    [
        "Data Scientist",
        "Python Developer",
        "Cyber Security",
        "AI Engineer",
        "Software Engineer"
    ]
)

# Generate Question
if st.button("Generate Question"):

    question, keywords = get_question(role)

    st.session_state.question = question

    st.session_state.keywords = keywords

# Show Question
if "question" in st.session_state:

    st.subheader("Interview Question")

    st.write(st.session_state.question)

    answer = st.text_area("Write Your Answer")

    # Analyze Answer
    if st.button("Analyze Answer"):

        score, matched = analyze_answer(
            answer,
            st.session_state.keywords
        )

        feedback = generate_feedback(score)

```

```
st.success(f"Score: {score}/10")

st.info(feedback)

st.write("✅ Matched Keywords:")

st.write(matched)
```

Overwriting app.py

✎ Import Ngrok

```
from pyngrok import ngrok
```

✎ Start Streamlit

```
!streamlit run app.py &>/dev/null&
```

```
ngrok.set_auth_token('YOUR_AUTH_TOKEN')
```

✎ Ngrok Account

```
from pyngrok import ngrok

ngrok.set_auth_token("3DTcGIggT0lENewqDrLJhyLoqoa_6urk2igV9urjc9PQtsZpi")
```

✎ Generate Website Link

```
public_url = ngrok.connect(8501)
print(public_url)
```

NgrokTunnel: "<https://flame-sister-strongly.ngrok-free.dev>" -> "<http://localhost:8501>"